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From Regional Disputes to Global Solutions: A Fairness-Based Game-Theoretic Model for Sustainable River Water Allocation

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Abstract
Conflicts from competing interests are inherent to human interactions, and water resource management, a vital yet finite asset, is no exception. Sustainable water sharing is crucial for equitable use and aligns with UN SDGs, particularly SDG 6 (clean water and sanitation), SDG 11 (sustainable cities), and SDG 17 (partnerships). This study applies the Fairness and Equity (F&E) technique to the Krishna River dispute among the states of Karnataka, Maharashtra, Telangana, and Andhra Pradesh. Key allocation factors include drainage area, virgin runoff, population, cultivable area, and water quality; the latter indirectly affects allocation through environmental penalties. Using a cooperative negotiation framework integrated with hydrological modeling, water shares are allocated as follows: Maharashtra (28%), Karnataka (41%), Telangana (17%), and Andhra Pradesh (14%). This distribution addresses the Krishna conflict while fostering sustainable practices and partnerships. Beyond this case, the study offers a replicable framework for transboundary river sharing that quantifies fairness, equity, and sustainability for future generations.

Keywords Fairness and equity · Water conflict resolution · Cooperative negotiation · Krishna river · Riparian states · Sustainable water management

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1 Introduction

Rivers are integral components of the hydrological cycle, sustaining ecological balance, agricultural productivity, and socioeconomic development. Beyond serving as surface water bodies, they contribute significantly to groundwater recharge through bank infiltration and riverbed seepage, thereby maintaining baseflow during dry seasons (Niu et al. 2022). This dual surface-subsurface interaction emphasizes the strategic importance of rivers in ensuring water security and ecological stability. However, the allocation of river water resources often triggers conflicts among riparian entities, threatening the sustainability of both surface and groundwater systems (Sohrabi et al. 2022).

In India's peninsular region, characterized by non-perennial rivers and high rainfall variability, inter-state water conflicts are recurrent (Angelakis et al. 2021). Seasonal monsoonal dependence intensifies disputes during dry years, as exemplified by the Krishna and Cauvery river basins (Ravikumar et al. 2021). Such conflicts extend beyond hydrological scarcity; they also reveal institutional, political, and socioeconomic asymmetries among states. These persistent challenges underscore the need for equitable, transparent, and data-driven frameworks for water sharing that align with sustainability principles (Zheng et al. 2022).

To address this global concern, water governance must incorporate the principles of the United Nations Sustainable Development Goals (SDGs)-notably SDG 6 (Clean Water and Sanitation), SDG 11 (Sustainable Cities and Communities), and SDG 17 (Partnerships for the Goals)—which emphasize fairness, inclusivity, and cooperation (Vaio et al. 2021; Arora And Mishra 2022). Achieving these targets requires embedding justice-oriented negotiation mechanisms into water-allocation decision-making (Rasinski 1987).

1.1 Evolution of Fairness-Based Allocation Frameworks

Transboundary water allocation frameworks have evolved from legal doctrines toward science-based cooperative mechanisms. Early doctrines such as the Riparian Doctrine, Harmon Doctrine, and Natural Flow Theory prioritized unilateral rights or downstream privileges, often disregarding hydrological interdependence (Kibler, 2014; Odendahl 2010). Contemporary frameworks emphasize hydrological equity, stakeholder participation, and sustainability (Wolf 1999; Wouters 2000; Valipour et al. 2024). Table 1 synthesizes historical and modern approaches to transboundary water allocation, illustrating this paradigm shift from rigid legalism to integrative governance. This table presents twenty allocation frameworks applied to transboundary water disputes, ranging from classical legal doctrines to contemporary methodologies. Analysis of these frameworks reveals that modern transboundary governance increasingly combines hydrological science, cooperative decision-making structures, and equity-based criteria. In contrast, historical approaches relied primarily on legal precedent and upstream advantage (Ma et al. 2024).

Fairness-based decision-making integrates three complementary justice dimensions: procedural, distributive, and interactive justice. Procedural justice ensures inclusive participation and transparency, while distributive justice concerns the equity of outcomes. Interactive justice, on the other hand, emphasizes mutual respect and recognition during negotiations. Together, these form the conceptual foundation for the Fairness and Equity (F&E) allocation model (Panday et al. 2021a), which transforms subjective fairness into quantifiable, cooperative outcomes.

Table 1 Historical and contemporary frameworks for transboundary water Allocation. Global frameworks, from classical legal doctrines to modern methods, illustrate a shift from precedent- and upstream-based rules to hydrology-informed, cooperative, and equity-driven governance

S. No.	Doctrine/Framework	Key Feature	Methodology	Application/ Example
1	Riparian Doctrine Wheeler et al. 2016	Riparian states have exclusive rights; Non-riparians are excluded	Comparative international law analysis	Mississippi River (Mexico-US)
2	Harmon Doctrine Kibler, 2014	Unilateral upstream use without downstream consideration	Historical treaty analysis	Rio Grande (US-Mexico)
3	Natural Flow Theory Odendahl 2010	Downstream riparian priority: Natural flow preservation	Natural flow doctrine application	Nile River (Egypt-Sudan)
4	Prior Appropriation Benson 2011; Wheeler et al. 2018	Prior use establishes allocation priority	Prior appropriation legal framework	Colorado River (US-Canada)
5	No Significant Harm Gupta And Schmeier 2020	No significant harm to other parties; Compromise required	No-harm principle & negotiation	Lake Lanoux (Spain-France)
6	Equitable Apportionment Wolf 1999; Wouters 2000	Equal opportunity; Equality in sharing principle	Equal opportunity & proportional allocation	Arkansas River (Kansas-Colorado)
7	Fairness & Equity (FE) Cooperative Negotiation ***[Present Study]***	Quantifiable fairness; Stakeholder-ranked factors (drainage area, runoff, population, cultivable area)	Cooperative game theory; Multi-criteria ranking; Iterative compromise algorithm	Krishna River (Karnataka 41%, Maharashtra 28%, Telangana 17%, AP 14%)
8	Transboundary Water Rights & Cooperation Mndzebele (2025***)***.	Conflict prevention through functional river basin institution-building	Institutional analysis; Policy review; Multi-stakeholder frameworks	Sub-Saharan Africa (SADC region)
9	Water Governance for Peacebuilding Abdelgalil and Ahmed (2025***)***.	Uses water governance as a mechanism for peacebuilding in conflict regions	Qualitative drivers analysis; Policy framework review	Sudan water conflicts
10	Conflict Exploration Framework Zaghiyan et al. (2025).	Conflict exploration for sustainable management of water development plans	Conflict analysis; Integrative assessment framework	Tigris & Euphrates Basin (Israel-Palestine)
11	Multi-Objective Water Allocation Liu et al. 2025	Multi-target decision framework for cascade reservoirs	Multi-target decision framework; Optimization	Yellow River Basin cascade system
12	Stakeholder-Driven Multi-criteria Decision Making Tang et al. (2025***)***.	States define their own equity-maximizing factors; Compromise levels (1–3)	Preference vectors; Factor-weighting; Allocation share maximization	Multi-state negotiation framework
13	Cooperative Game Theory Application Pournabi et al. (2025).	Cooperative game theory to develop water management strategies	Game-theoretic modeling; Water management optimization	Hour Al-Azim wetland restoration
14	Transboundary River Basin Institution-Building Varma et al. (2025).	Institutional frameworks for multi-stakeholder cooperation	Institutional design; Policy integration	Brahmaputra River basin
15	Intergovernmental Cooperation in Transboundary Systems Xia and Bao (2025***)***.	Evolution and mechanism of intergovernmental cooperation	Institutional analysis; Cooperation mechanisms	Taihu Basin, China

Table 1 (continued)

S. No.	Doctrine/Framework	Key Feature	Methodology	Application/ Example
16	Sustaining Transboundary Cooperation Zhang et al. (2025).	Pathways for sustaining stable transboundary co-operation under changing hydrology	Adaptive cooperation framework; Hydro-logical scenarios	Lancang-Mekong River Basin
17	Integrated Water-Energy-Sediment-Ecology Management Liu et al. (2025).	Multi-objective optimiza-tion of competing water demands	Multi-target framework; Systems optimization	Yellow River mainstream cascade
18	Climate-Adaptive Water Allocation Benti et al. (2025).	Water allocation under hydrological variability & climate change	Climate scenario modeling; Adaptive management	Multiple basins under future climate
19	Water-Energy-Food Nexus Coordination Zheng et al. (2022)	Water-energy-food nexus coupling & coordination	Hybrid framework; Spatial heterogeneity analysis	Yellow River Basin
20	Water Quantity-Quality Integrated Management Han et al. (2025).	Water carrying capacity & sustainable use limits	Entropy weight + CRITIC methods; Compre-hensive evaluation	Dadu River Basin

1.2 The Krishna River Dispute: Context and Challenges

The Krishna River-originating near Mahabaleshwar in Maharashtra-extends over 1,400 km before joining the Bay of Bengal and sustains nearly 8% of India's land area (Chanapathi And Thatikonda 2020). The river traverses Maharashtra, Karnataka, Telangana, and Andhra Pradesh, supporting agriculture, industry, and domestic supply for millions of people. The 2015 bifurcation of Andhra Pradesh (Pullaiah 2015) introduced new administrative boundaries, intensifying inter-state allocation disputes. The Krishna Water Disputes Tribunals (KWDT I, 1973; KWDT II, 2010) established allocations based on historic hydrology, but these static entitlements have struggled to accommodate climatic variability, infrastructure expansion, and emerging environmental constraints.

Climatologically, the basin exhibits stark heterogeneity, with rainfall exceeding 1,000 mm yr^{-1} in the Western Ghats but declining below 500 mm yr^{-1} in interior dry zones (Bouwer et al. 2006). This spatio-temporal asymmetry complicates uniform water-sharing arrangements. As anthropogenic pressures increase, maintaining hydrological balance, agricultural resilience, and ecological integrity requires models that combine scientific rigor with cooperative legitimacy.

Existing approaches—including game-theoretic, optimization, and rule-based models—have enhanced the analytical rigor of water-sharing studies; however, they often treat allocation as a zero-sum problem or overlook participatory fairness. Models by Ravikumar and Khosa (2007); Panday et al. 2021b. advanced cooperative and metagame perspectives yet lacked integrated mechanisms to incorporate multi-criteria stakeholder ranking, temporal variability, or water-quality accountability.

To address these limitations, this study proposes a Fairness and Equity (F&E) cooperative negotiation framework that quantifies fairness through stakeholder-defined rankings of hydrological and socioeconomic factors. Each riparian State prioritizes drainage area, cultivable area, virgin runoff, and population, representing distinct equity dimensions—spatial,

agricultural, hydrological, and demographic. The model iteratively computes allocations until the total equals 100%, ensuring convergence toward a feasible, collectively acceptable solution.

The approach uniquely couples this negotiation algorithm with ArcSWAT-based runoff simulation, providing physically validated hydrological data to support fair negotiations. By embedding the simulation within a cooperative game context, the model bridges the gap between hydrological modeling and stakeholder decision-making. This linkage has seldom been operationalized in earlier studies.

Furthermore, the framework incorporates a water-quality penalty mechanism, whereby upstream pollution (quantified by Biochemical Oxygen Demand, BOD) results in compensatory payments to downstream states. This innovation bridges the gap between quantity and quality, fostering accountability while preserving negotiated allocations. Together, these elements enable an adaptive, evidence-based system that balances equity, efficiency, and environmental responsibility.

The present work contributes a three-tiered innovation to the field of transboundary water governance. First, it quantifies fairness through a cooperative algorithm that translates procedural and distributive justice into measurable allocation outcomes based on stakeholder preferences, thereby operationalizing abstract fairness principles within a replicable computational structure. Second, it integrates hydrological validation using ArcSWAT simulations to ensure physical consistency of modelled allocations with observed river discharge, establishing a direct linkage between hydrological realism and negotiation dynamics. Third, it introduces a BOD-linked compensation mechanism that connects water allocation with environmental performance, enabling accountability for water-quality deterioration without altering negotiated shares. Collectively, these components advance transboundary allocation research by combining hydrological modeling, cooperative negotiations, and environmental responsibility within a unified decision-support framework.

This integrated framework tested on the Krishna River basin, advances beyond prior studies by unifying fairness principles, physical modeling, and ecological accountability within a replicable decision-support system. Its application demonstrates how quantitative fairness modeling can transform adversarial negotiations into transparent, cooperative water diplomacy, contributing directly to the operationalization of global sustainability goals (SDG 6, 11, 17).

2 Materials and Methods

2.1 Study Area

The study area focuses on the Krishna River, a prominent river system in peninsular India. The Krishna River holds significant geographical and ecological importance, covering approximately 8% of the region's total geographical area (Fig. 1). It is one of the major rivers in India, playing a crucial role in supporting the livelihoods of millions of people residing in the riparian states of Andhra Pradesh, Telangana, Maharashtra, and Karnataka (Kumar And Jayakumar 2020). The Krishna River originates in the Western Ghats range near Mahabaleshwar in Maharashtra and flows eastward for about 1,400 km

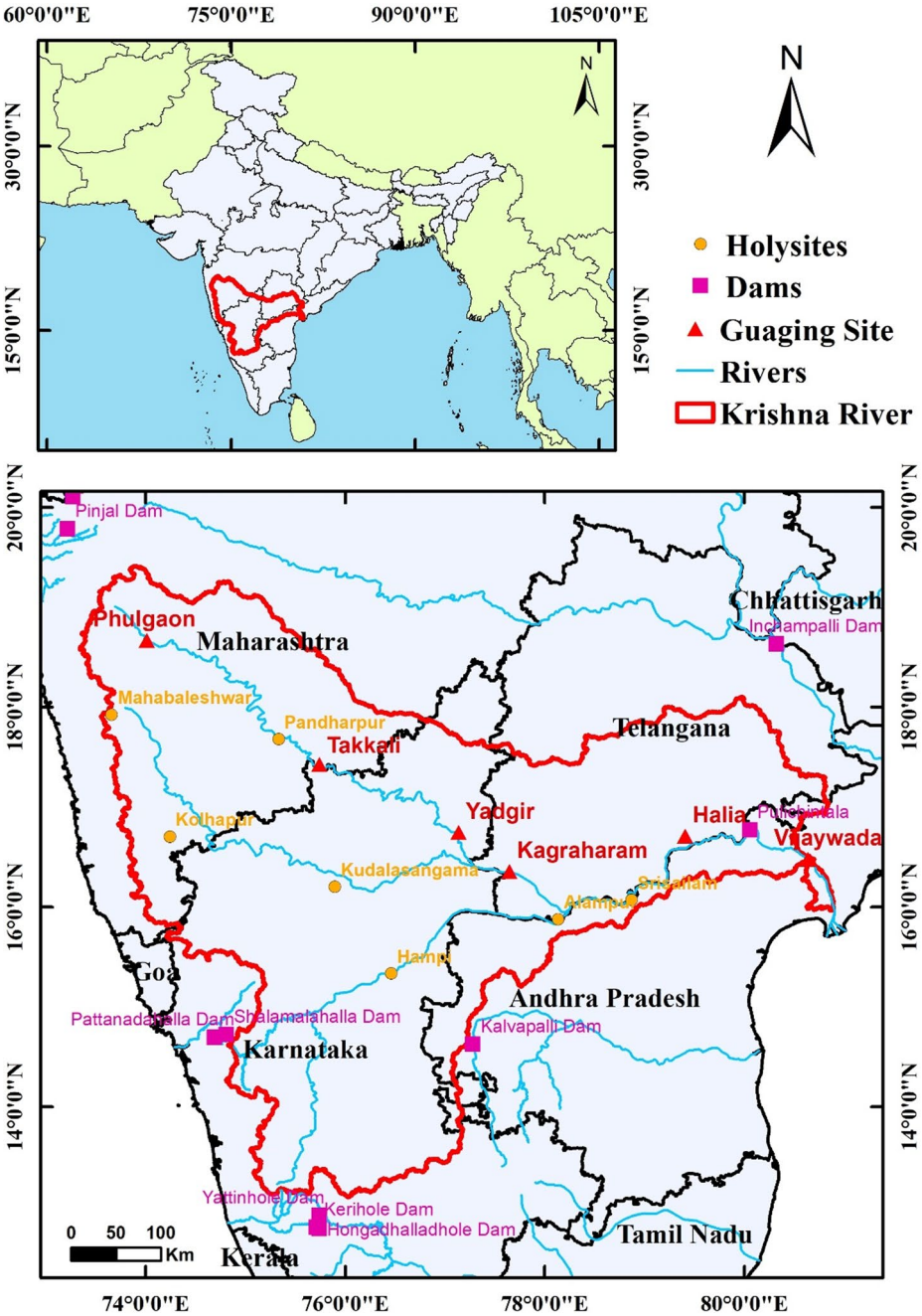

Fig. 1 Study Area. The Krishna River basin, showing the areal coverage of the riparian states, i.e., Maharashtra, Karnataka, Telangana, and Andhra Pradesh. The Krishna River originates in Maharashtra (the uppermost riparian State). It forms a delta in Andhra Pradesh (the lowermost riparian State), along with critical gauging sites, holy sites, and significant dams

before joining the Bay of Bengal (Chanapathi And Thatikonda 2020). It traverses diverse landscapes, including hilly terrains, plateaus, and fertile plains, providing a wide range of ecosystem services and supporting a rich biodiversity. The river system has several tributaries, including the Tungabhadra, Bhima, and Ghataprabha rivers, which contribute to its water flow and overall hydrological dynamics. The Krishna River basin, as depicted in Fig. 1, encompasses a vast area and is crucial for various sectors, including agriculture, industry, hydropower generation, and domestic water supply (KWDT 2010; Prasad And Patil 2008).

Climatically, the Krishna River basin exhibits considerable variability, ranging from tropical monsoon conditions in the coastal and southern regions to semi-arid and arid zones in the northern interior. The monsoon season, extending from June to September, accounts for the bulk of annual precipitation, while the non-monsoon months are predominantly dry. This seasonal disparity results in pronounced fluctuations in river discharge, necessitating the implementation of careful water management strategies. Annual rainfall varies significantly across the basin, from over 1,000 mm in the Western Ghats to less than 500 mm in the rain-shadow areas. Such heterogeneity in climate contributes to both spatial and temporal challenges in equitable water allocation among the riparian states (Bouwer et al. 2006).

To conduct the study on the Krishna River, comprehensive data is gathered and analyzed, including information related to the river's hydrology, water quality, land use, population distribution, and socioeconomic factors. These data provide a holistic understanding of the river system's characteristics and dynamics.

Understanding the complexities of water sharing and allocation within the Krishna River basin is crucial for addressing the socioeconomic and environmental challenges faced by the riparian states. The study aims to develop a comprehensive framework that promotes fairness, equity, and optimal utilization of the Krishna River's water resources. By considering the diverse aspects of the river system, the study aims to contribute to sustainable water management practices, ensuring the long-term well-being of both humans and the environment.

2.2 Data Used

A comprehensive set of spatial, hydrological, climatic, and socioeconomic data was used to characterize the Krishna River basin and support runoff simulation and allocation analysis. Digital Elevation Model (DEM), soil cover, and land use/land cover data were obtained from publicly available remote sensing sources. Climatic data (precipitation and temperature) were sourced from the India Meteorological Department (IMD), while discharge records came from the Central Water Commission (CWC). Demographic and agricultural datasets, including population and cultivable area statistics, were extracted from official census and government reports to support fairness-based water allocation modelling.

The DEM was used to delineate drainage boundaries and identify the contributing catchment area for each State, quantifying their hydrological footprint in the basin. Cultivable area data represent the primary water-dependent land use—agriculture—and directly influence irrigation demand, the dominant sector driving the dispute. Virgin runoff data quantify

the natural water contribution of each State, excluding anthropogenic modifications, thereby ensuring that upstream hydrological equity is considered. Population data from the 2011 Indian census serve as a proxy for domestic and economic water needs while maintaining temporal consistency to avoid incentivizing artificial growth. These factors are integrated into the cooperative negotiation framework by allowing riparian states to rank them according to their strategic interests. The model uses these rankings to compute weighted contributions and iteratively resolve allocations through compromise, ensuring that no single metric dominates and the outcome reflects both procedural and distributive fairness. Thus, the chosen data types not only anchor the model in empirical evidence but also ensure legitimacy in the negotiation process.

2.3 Cooperated Negotiations

The proposed methodology is grounded in a justice-centered framework with three dimensions—procedural, distributive, and interactive justice—each of which is explicitly embedded in the water-sharing process. Procedural justice is ensured through participatory negotiation, where each riparian State independently defines and ranks allocation factors (e.g., drainage area, runoff contribution) according to its perceived needs, thereby embedding fairness directly into the decision-making process. Distributive justice is achieved by translating these rankings into water shares that reflect relative entitlements, ensuring each State's allocation is proportional to agreed-upon hydrological, geographical, and demographic criteria. Interactive justice emerges from an iterative negotiation process in which states engage transparently, acknowledge each other's constraints, mutually recognize competing demands, and converge through compromise toward a feasible solution. Together, this tripartite justice framework underpins the cooperative allocation strategy, ensuring that both the process (procedural and interactive justice) and the outcome (distributive justice) are perceived as fair and equitable. In this study, the methodology involved simulating and validating the runoff of the Krishna River basin using the ArcSWAT model (Anand et al. 2018), which is based on the water budget equation and balances key components of the hydrological cycle, including precipitation, surface runoff, evapotranspiration, water seepage, return flow, and soil water content.

The ranked preference matrix **(**Table 2**)** reveals distinct prioritization patterns among the four riparian states, reflecting their hydrological, demographic, and agricultural characteristics.

Table 2 The preference vector shows the ranking of each factor based on its decreasing contribution (in percentage). Factors 1, 2, 3, and 4 represent cultivable area, drainage area, Virgin runoff contribution, and population, respectively

State	Rank 1	Rank 2	Rank 3	Rank 4
Maharashtra	Virgin Runoff (33%)	Population (31%)	Cultivable area (26.79%)	Drainage Area (21%)
Karnataka	Drainage Area (48%)	Cultivable area (43.85%)	Virgin Runoff (39%)	Population (35%)
Andhra Pradesh	Cultivable Area d(20.11%)	Drainage Area (12%)	Virgin Runoff (11%)	Population (11%)
Telangana	Population (23%)	Drainage Area (19%)	Virgin Runoff (17%)	Cultivable area (9.25%)

3 Results and Discussions

3.1 SWAT Analysis

The Soil and Water Assessment Tool (SWAT) models were used to assess runoff at the Takli and Halia sites (locations shown in Fig. 1), with the results illustrated in Figs. 2 and 3, respectively. These figures present observed and simulated runoff along with the coefficient of determination (R^2), which indicates the correlation between observed and simulated discharge. For the Takli site (Fig. 2), manual calibration involved a judgment-based, trial-and-error adjustment of parameters to achieve accurate simulations, yielding an R^2 value of 0.75, which indicates a reasonably strong correlation and a reliable representation of runoff dynamics. For the Halia site (Fig. 3), similar calibration procedures were applied, and the validation phase produced an R^2 value of 0.85, indicating a strong correlation between observed and simulated discharge and effective capture of runoff characteristics.

The high R^2 values at both sites (0.75 for Takli and 0.85 for Halia) indicate that the SWAT model accurately reproduces the observed discharge patterns, validating the effectiveness of the manual calibration process. Aligning simulated discharge with observed data enhanced the model's ability to simulate runoff accurately, improved understanding of hydrological dynamics at both sites, and supported the applicability of the SWAT model for future studies and water-resource management decisions in these areas.

3.2 Factors for Water Allocation

In the water allocation process, selecting appropriate factors is crucial, as they reflect the aspirations and interests of all parties. The United Nations Convention on the Law of the Non-Navigational Uses of International Watercourses (UNCIW) provides guidelines for determin-

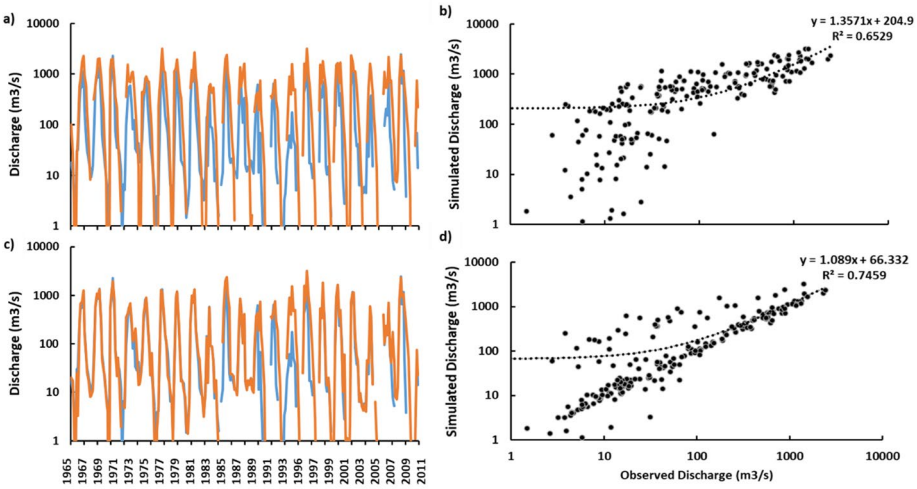

Fig. 2 Observed and simulated monthly discharge at the Takli gauging station (Calibration run; $N = 378$ months). The relationship between observed and simulated discharge follows a logarithmic trend, expressed as: $\ln(Q_{sim}) = 0.66 \times \ln(Q_{obs}) + 185.77$. Model performance statistics are: $R^2 = 0.75$; NSE = 0.871, and KGE = 0.535,

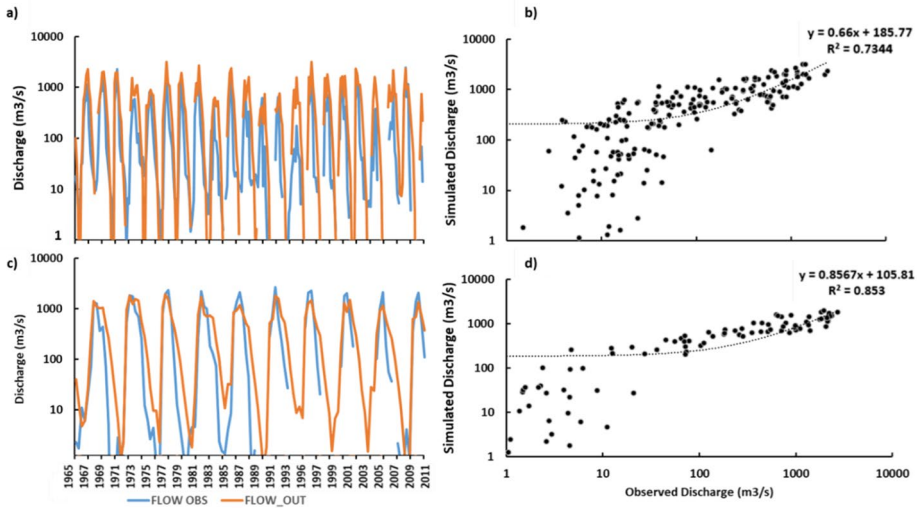

Fig. 3 Observed and simulated monthly discharge at the Halia gauging station (Validation run; $N = 378$ months) using an independent parameter set. The relationship between observed and simulated discharge also follows a logarithmic trend, represented as: $\ln(Q_{sim}) = 0.8567 \times \ln(Q_{obs}) + 105.81$; $R^2 = 0.857$. Model performance statistics are: NSE = 0.723, KGE = 0.708, and PBIAS = −10.88%

ing these factors, which should cover drainage, climate, and demographic aspects (Jung 2024; Singh et al. 2024). Each player in the dispute chooses the factor that maximizes its share of the total. For Krishna River water allocation, the selected factors include both water quantity (drainage area, cultivable area (CA), virgin runoff, population) and water quality aspects.

The allocation of Krishna River waters considers four fundamental factors—drainage area, cultivable area, virgin runoff, and population—each representing a distinct dimension of fairness and equity among riparian states. Drainage area captures the geographic extent and hydrological influence of each State, recognizing spatial contributions and ecological responsibilities. Cultivable area reflects the basin's agricultural dependency, aligning allocations with irrigation needs and supporting equitable access to water for food production and rural livelihoods (KWDT 2010). Virgin runoff, defined as the natural surface flow unaffected by anthropogenic alterations, promotes procedural justice by neutralizing advantages from differing economic and infrastructural capacities, thus ensuring a level playing field. The population represents socioeconomic demand and anchors distributive fairness by linking water entitlements to human needs and dependency.

To avoid distortions, 2011 census data are used, with Telangana's population derived from districts within the Krishna basin. Together, these factors create a balanced, transparent framework that integrates hydrological, agricultural, environmental, and social dimensions, ensuring that water sharing in the Krishna River basin is equitable, scientifically informed, and aligned with sustainable resource governance.

3.3 Cooperative Framework for Negotiations and Fair Allocation

The cooperative framework for negotiations (Kyriazi, 2017) establishes a collaborative approach for fair and equitable allocation of Krishna River water among riparian states. The

framework is based on several key assumptions. First, the total water allocation is capped at 100% and distributed among states on a percentage basis, ensuring that allocations remain within available resources and preventing overallocation. Second, the parties recognize the need for compromise when the total allocation based on their preferences exceeds 100%, implying an infeasible outcome. Third, all parties agree that the process must be fair and equitable. Each factor in the allocation procedure is assigned equal weight, reflecting its importance for each State's water requirements and preventing any factor from being disregarded or undervalued.

Water allocation is determined by allowing each riparian State to rank the factors according to their perceived contribution. For example, Karnataka assigns the highest contribution to factor 2 (drainage area), followed by factor 1 (cultivable area), factor 3 (virgin runoff), and factor 4 (population). The allocation process then employs a compromise mechanism, in which each State's top-ranked factors are considered in a negotiated setting. If the resulting total allocation exceeds 100%, indicating infeasibility, states progressively consider their second-ranked factors and continue this iterative process until a feasible solution is obtained. This procedure aims to foster collaboration, shared responsibility, and an inclusive, sustainable allocation of Krishna River water, while ensuring that the interests and needs of each State are explicitly considered.

The bifurcation of Andhra Pradesh and the creation of Telangana introduced a new stakeholder, intensifying the Krishna River water distribution conflict. Telangana has accused Andhra Pradesh of deliberately neglecting irrigation projects in its region and providing inadequate budget allocations (Panday et al. 2021a, 2021b). These grievances must be addressed, particularly because the contested projects involve large-scale hydraulic infrastructure—dams, canals, and barrages—that, unlike an automobile plant, cannot be physically relocated between states (Chokkakula 2016).

To achieve a fair and equitable allocation of water resources, a Framework for Negotiations and Fair Allocation (F&E allocation) has been derived and implemented. Under this framework, each State is free to propose its own set of factors that maximizes its share of water, recognizing that direct involvement of the parties in conflict resolution enhances the likelihood of successful outcomes. The proposed methodology addresses equity and fairness by employing proportional allocations for distributive justice, ensuring that each State's share of Krishna River water reflects its relative contributions and needs. The cooperative–equal opportunity negotiation model embedded in this methodology targets procedural justice by explicitly incorporating the perceptions and interests of all parties.

Figure 4 presents the relative contribution of the four factors considered for each riparian State in the Krishna River conflict. This visual representation clarifies the significance of each factor and its influence on the allocation process. By examining these contributions, decision-makers can gain a deeper understanding of allocation dynamics and make informed choices regarding the equitable distribution of water among the states. Implementing the F&E allocation framework, which allows each State to emphasize factors that maximize its share, aims to ensure fairness, equity, and sustainability in Krishna River water allocation. This approach also addresses the inherent complexities of transboundary water distribution through a cooperative and inclusive process.

According to the procedure outlined in Ravikumar and Khosa (2007), if each State insists on allocations proportional to its highest-ranked factor, the total water requirement becomes 124.11% (33 + 48 + 20.11 + 23 = 124.11%), which exceeds the available resources and is

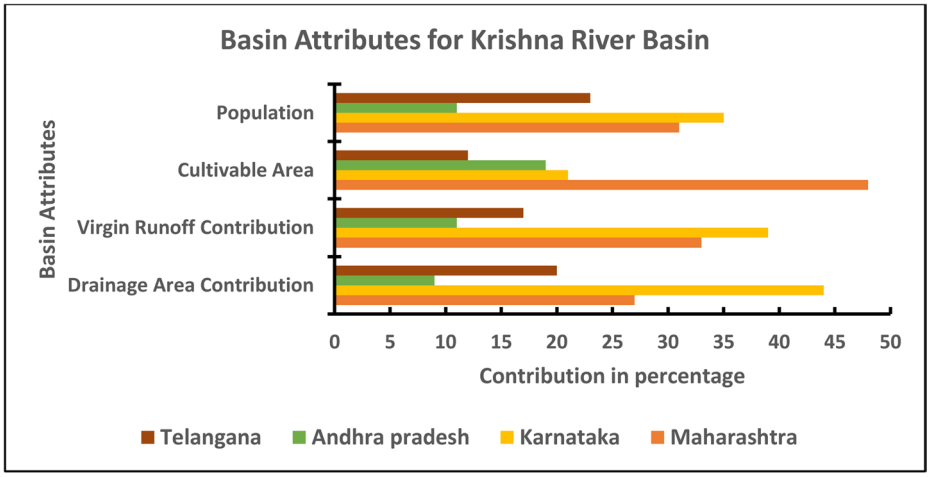

Fig. 4 Final allocation factors. Plot showing the factors of allocation and their relative contribution (in percentage terms). Each State has the highest allocation percentage in different categories. This highlights the power of negotiated cooperation, in that each player is free to put forth its own best options

therefore infeasible. Consequently, each State must compromise to reach a feasible solution. As a first level of compromise, the states consider their top two contributions as the basis for allocation, incorporating their second-ranked factor into the calculation of water shares. This iterative process continues—progressively including lower-ranked factors—until the total allocation converges to exactly 100%.

A third level of compromise is attained when all selected factors are considered for each State, resulting in a feasible allocation where the total share sums to 100%; for example, (27.95 + 41.46 + 13.53 + 17.06) = 100%. Table 3 summarizes this procedure, presenting successive levels of compromise and the corresponding allocations. Supplementary item SI 4-Fig. 2, and SI provide a visual representation of the final distribution of shares among the ripar-

Table 3 Cooperative negotiations analysis for the fair and equitable share allocation. Final allocation is reached when the final allocation percentage reaches 100%. Levels 1 and 2 show the compromise in case the total allocation exceeds 100%. Level 3 emphasizes that all parties involved recognize the necessity of compromise to achieve a lasting outcome

Allocation considering Rank 1	(33 + 48 + 20.11 + 23) = 124.11	> 100%	Infeasible
Compromise Level 1 (Rank1; Rank2)	(32 + 45.93 + 16.05 + 21) = 114.98	> 100%	Infeasible
Compromise Level 2 (Rank1; Rank2; Rank3)	(30.26 + 43.62 + 14.37 + 19.67) = 107.92	= 107.9%	Infeasible Solution
Compromise Level 3 (Rank1; Rank2; Rank3; Rank4)	(27.95 + 41.46 + 13.53 + 17.06) = 100	= 100%	Feasible Solution

ian states. By following this structured procedure within a cooperative negotiation setting, the states can arrive at a mutually acceptable allocation that reflects their respective contributions and priorities, thereby promoting fairness, cooperation, and sustainability in the Krishna River water-sharing arrangement. The detailed step-by-step algorithm of the F&E cooperative negotiation procedure, along with a worked numerical example, is provided in Supplementary Items SI5 and SI6.

Recent studies, such as those by Johnson and Zhang (2025) and Elias (2024), highlight the importance of participatory frameworks and ecosystem restoration in managing transboundary water systems. However, they often lack integration with hydrological modeling or do not provide quantitative mechanisms for fairness-based allocations. Similarly, Ravikumar et al. (2021) extended metagame analysis to the Cauvery dispute by incorporating interannual runoff variability; however, our study advances this work by embedding cooperative negotiations directly within a runoff-validated allocation framework using ArcSWAT. Furthermore Panday et al. (2021a, 2021b) applied game-theoretic models to the Krishna conflict but did not include water quality penalties or allow riparian players to define their own equity-maximizing criteria, as enabled in our F&E framework. This multi-criteria, stakeholder-driven, and temporally robust model distinguishes our study and enhances its policy relevance for replicable river basin governance.

The cooperative negotiation process allows states to actively determine the factors that maximize their water allocation, thereby fostering ownership of the agreed solutions and increasing the likelihood of successful implementation. This approach recognizes that the states themselves are responsible for achieving a fair and equitable agreement. To address temporal variability in runoff and its implications for sustainability, the analysis incorporates the dynamic factor 3, virgin runoff potential, evaluated at four levels of dependability: Max. Flow (MAX), Upper Quartile Flow (UQ), Median Flow (MED), and Lower Quartile Flow (LQ). Fair and equitable (F&E) analyses are conducted for each of these flow categories. The possibility of non-homogeneity in runoff availability across the basin is also considered: F&E allocations are derived both for cases where runoff contributions in all states fall within the same flow category and for instances where runoff availability differs across categories.

Table 2 (***SI 5***-Table 2) summarizes the results of the temporal variation analysis. The methodology follows that used in Tables 2 and 3, though numerical results may differ slightly. The findings show that the adopted method is not significantly affected by spatio-temporal variations in flow. A key implication is that allocations should be made on a percentage basis rather than in absolute volumes. During low-flow periods, all states must recognize the need for compromise and share the available resources judiciously.

Overall, the cooperative negotiation approach is adaptable to temporal variations in runoff while maintaining fair and equitable allocation. By accounting for multiple flow categories and potential non-homogeneity in runoff availability, the analysis reflects the dynamic nature of water resources. It highlights the need for cooperation and compromise among riparian states. If all states commit to the negotiated solution and adhere to the agreed allocations, the framework has the potential to provide a durable settlement. By ensuring fairness and equity in water distribution and promoting the sustainable use of the shared resource, this approach can contribute to regional peace and support the long-term preservation and harmonious management of the Krishna River's water resources.

In this framework, the "opinions" of the stakeholders correspond to the best choices already tabled by the four riparian States in the official records of KWDT-I and KWDT-II. The options

we use, therefore, reflect positions that each State has formally advanced as desirable or acceptable during prior adjudication. However, any arbitrary demands that lack a measurable basis are not considered, since the present framework is explicitly grounded in an objective and quantifiable approach. Only factors for which consistent basin-level data are available—drainage area, cultivable area, virgin runoff, population, and water quality (through penalties/compensation)—are retained. The ranking of factors is also derived objectively. For each State, its contribution to every factor is computed as a relative percentage, and factors are then ranked according to these percentage contributions. This ensures that stakeholder preferences are incorporated into the model through documented, data-supported choices, while the allocation remains grounded in transparent, reproducible, and non-arbitrary criteria.

3.4 Water Quality Factors

Incorporating water quality into the allocation of shared water resources is essential to ensure that both quantitative and qualitative aspects of river management are addressed equitably. Downstream states play a critical ecological role by conveying floodwaters to the sea and exporting salts and dissolved pollutants. In contrast, upstream states face risks of submergence if reservoir releases are delayed during floods. However, downstream riparian areas bear the significant impacts of agricultural runoff and industrial effluents originating upstream. It is therefore imperative to hold polluting states accountable and impose penalties for the deterioration of shared water quality. Two practical mechanisms may be adopted: (i) installation and operation of treatment facilities by upstream states to meet agreed water quality standards, and (ii) monetary or in-kind compensation to affected downstream states based on negotiated costs or the expenditure required for standard cleaning procedures (Agarwal et al. 2024). Although water quality negatively influences allocation outcomes, actual volumetric shares cannot be reduced solely on this basis, given the scarcity of the resource; compensation provides a fairer and more implementable alternative. Industrial development, while essential for economic growth, often compromises water quality; restoration, reuse, and sustainable utilization must therefore be prioritized for this limited, shared resource.

Once quantitative allocations are determined, qualitative concerns must be addressed. A key challenge is the lack of comprehensive and openly shared water quality data. Unlike factors such as drainage area or runoff contribution, which positively affect allocations, deteriorated water quality contributes negatively. Meeting merely the permissible limits is insufficient, as downstream states may still be burdened. Instead, the percentage deterioration of a selected parameter between entry and exit points within a state provides a more meaningful basis for penalties. In the context of Telangana and Andhra Pradesh, compensation may also be offered in-kind through agricultural produce, using the fertile delta of Andhra Pradesh to balance resource constraints in Telangana. Details of the penalty computation framework are provided in Supplementary Item SI-6.

Implementation of the water-quality penalty is feasible within the existing monitoring structure. Each riparian State should maintain automatic and manual Biochemical Oxygen Demand (BOD) monitoring stations at its inlet and outlet sites, with a uniform sampling frequency (e.g., monthly). Median or 90th percentile BOD values over each adjudication period should be used to minimize short-term manipulation. Data verification would be undertaken jointly by a River Basin Authority or a Joint Monitoring Committee, with time-stamped

records stored in a centralized, accessible database. Annual independent laboratory audits and random verification sampling would supplement this. At the end of each monitoring window, penalties are computed automatically using predefined formulas, and compensations are processed through an escrow account managed by the Basin Authority. An appeals mechanism should allow re-testing of disputed samples. Enforcement can be strengthened by linking penalties to annual allocation orders and permitting partial fulfillment via verified pollution-control or treatment investments. Collectively, these measures render the system practical, transparent, and enforceable without new institutional arrangements.

For illustration, considering Chemical Oxygen Demand (COD) with a permissible limit of 25 mg/L, if COD rises from 10 to 20 mg/L within a drainage area, the downstream State cannot allow an increase exceeding 5 mg/L; the corresponding percentage deterioration is therefore critical. Only anthropogenic contributions are penalized, as natural variability lies beyond state control. Key water-quality parameters include pH, BOD, Dissolved Oxygen (DO), Total Coliform, and Fecal Coliform. Figure 5 illustrates the spatial variation of BOD_5 (mg/L) along the Krishna River, with progressive degradation from the upper to the lower riparian states. As identified by the Central Pollution Control Board (CPCB), BOD and fecal coliforms are the primary determinants of the water quality in the Krishna River. Elevated BOD corresponds to reduced DO and higher organic pollution, whereas low BOD indicates better oxygenation and lower microbial activity. This study, therefore, emphasizes BOD as the primary indicator for assessing water quality across riparian stretches, revealing that nearly all gauging sites exceed prescribed standards. Supplementary item SI-7 details the algorithm for the water quality-based penalty as applied in the present framework.

The proposed F&E framework is intended to operate within a realistic institutional setting. Implementation should be entrusted to a River Basin Authority or Joint River Commission with representation from all four riparian States and the Union government. This body would compile and periodically update the objective allocation factors (drainage area, cultivable area, virgin runoff, and population) and maintain a shared database of hydrological and water-quality data. For the water-quality penalty, each State would operate inlet and outlet BOD monitoring stations following standardised protocols, with joint verification and periodic independent audits to ensure credibility. At the end of each adjudication period, the Authority would apply the agreed formulas to compute penalties and compensations, settle these through an escrow account linked to annual allocation orders, and provide an appeals channel for contested measurements, including the option to offset part of the liability through verified investments in pollution-control infrastructure.

4 Conclusion

Transboundary rivers are increasingly at the center of disputes over limited and unevenly distributed water resources. This study proposes a structured, transparent Fairness and Equity (F&E) cooperative negotiation framework that provides a scientific and participatory basis for resolving such conflicts. By embedding procedural, distributive, and interactive justice, the approach supports inclusive decision-making, equitable outcomes, and respectful interaction among riparian states. Allowing stakeholders to prioritise allocation criteria according to regional needs, while maintaining a straightforward computational process, helps build legitimacy and trust in the results.

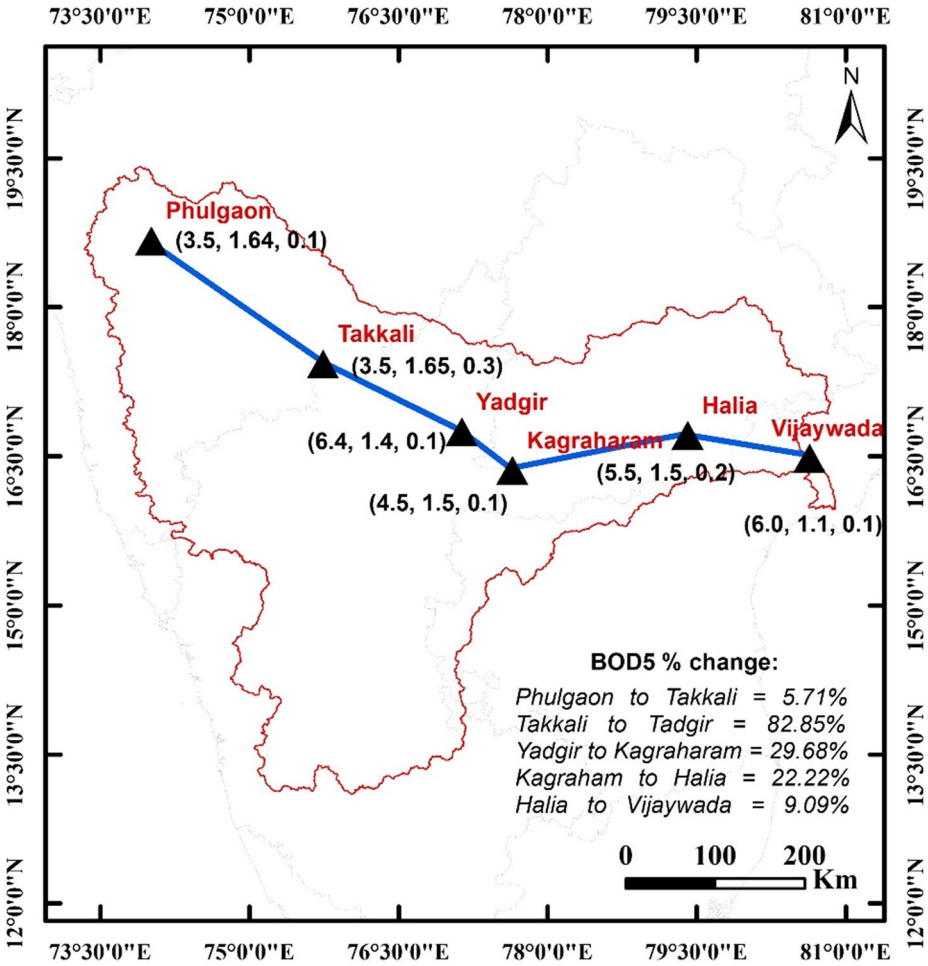

Fig. 5 Spatial variation of water quality parameter BOD_5. The plot shows the spatial variation of the BOD_5 (in mg/l) values from the initial riparian State to the last riparian State. The values in the brackets correspond to the minimum, average, and maximum values, respectively, and the percentage change is worked out for the maximum BOD_5

A distinctive feature of the framework is the explicit integration of water-quality considerations into allocation. A BOD-linked monetary penalty links water use with environmental responsibility, requiring states that deteriorate water quality to compensate affected downstream users. This connection between quantity and quality reinforces accountability without altering agreed volumetric shares and can encourage investment in pollution control and more careful resource use. The current F&E framework approaches fairness principles in a quantified, data-driven manner; it can be adapted to other transboundary basins with varying hydrological, institutional, or socioeconomic contexts. By aligning hydrological modelling with justice concepts and cooperative negotiation, it offers a practical tool to support more peaceful, balanced, and environmentally responsible river-basin governance.

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Declarations

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